



COM-480 - Data Visualization

BeanMap – Milestone 3

Process Book

Submitted By:

Victor Nahoul

Massimo Berardi

Noam Ifergan

Live demo: beanmap.vercel.app

Repository: github.com/com-480-data-visualization/BeanMap

1 Discovery & Narrative

Whether you are a dedicated coffee nerd or simply curious about the mechanics of macro-scale money-making, **BeanMap** invites you on an adventure into one of the most traded commodities in the world. Most of us drink coffee every day, yet we remain often unaware of where the economic value is actually extracted along the bean’s journey from a plantation in Ethiopia or Brazil to the espresso cup on a Lausanne café terrace.

Our initial vision was relatively simple: a flat world map decorated with arrows showing traded coffee quantities. But the moment we started surveying the landscape of existing visualizations, that idea collapsed. We found beautiful dashboards that displayed enormous tonnage flows from the equatorial belt to the Global North, and dense academic charts that buried the same information under a wall of acronyms. *None* of them answered the question that kept nagging us:

Where is the money actually made in the coffee supply chain?

The bean leaves Brazil as a raw commodity worth a few dollars per kilogram. By the time it is sold as a bag of roasted coffee in Rome, that same mass of coffee can be worth several times more. That value gap is the story, and it is a *geographic* story, because the gap is captured by a very specific set of countries that do not necessarily grow any coffee themselves. We set out to build a tool that would put that geography on a map for the first time, in a way that is rigorous, interactive, and genuinely beautiful to use.

Target audience. We aimed BeanMap at two overlapping personas. The first is the curious general reader: someone who drinks coffee, vaguely senses that “coffee is exploitative”, and wants to be told a story rather than handed a spreadsheet. The second is the data-literate reader (a journalist, an economics student, a development researcher) who already knows the broad strokes and wants the dashboard to let them *check* their intuition against forty years of data with a few mouse drags. Every design choice in the rest of this document is an attempt to satisfy both readers on a single page.

2 Data Wrangling & Methodology

The path to a working dataset was paved with detours. Coffee-specific commercial datasets turned out to be either too granular (price evolution of individual varieties) or too aggregated (data grouped by massive continental regions that erased exactly the granularity we needed: if Italy is folded into “Europe”, the entire story disappears). We then chased a professional bilateral-trade dataset that lived behind a paywall, before finally settling on the FAO (Food and Agriculture Organization of the United Nations) detailed trade matrix, which is free, gives country-pair-level quantities *and* values for every year from 1986 to 2024, and differentiates raw and processed coffee.

Data quality. Before committing, we did a fulfilment-rate audit in the EDA notebook (`EDA.ipynb`) and verified that the source-quality flag (“A” for official, “E” for estimated, “X” for external, “I” for imputed) is dominated by official reporting: unofficial sources combined account for at most 1.3% of the data. This gave us confidence to drop the source flag and aggregate freely.

The two items that matter. Coffee trade in the dataset is split across many product codes, but our entire story hinges on just two:

- Item 656: *Green Bean* (raw, unprocessed coffee).
- Item 657: *Processed Coffee* (everything from roasted whole beans through soluble coffee).

Every quantitative claim in BeanMap eventually reduces to a pair of numbers per country per year drawn from these two items.

Defining the Transformation Index. The metric at the heart of the project is the Transformation Index (TI), computed per country per year as

$$TI = \text{Raw Imports} - \text{Raw Exports}$$

A strongly *positive* TI marks a processing hub that buys green beans, transforms them, and eventually re-exports a higher-value product. A strongly *negative* TI marks a primary producer shipping beans out raw, keeping very little of the value chain at home, and eventually imports processed coffee. USA sits near the top; Brazil sits at the bottom. We chose this very simple formula after rejecting an earlier ratio-based draft for two reasons. First, ratios are hostile to small countries with low absolute volumes: a microstate that re-exports two containers of roasted coffee would appear as a 100% processor and dominate the map. Second, the difference formula has a clean physical interpretation. It is the net tonnage of raw coffee that physically disappeared inside the

country's borders, presumably to be processed for the domestic market or then exported. That makes the index a *mass-balance* quantity, and it lets the Sankey diagram conserve flow honestly (see §4.3). The only drawback of this index relies on the data source itself: we do not know how much coffee has been produced, processed and consumed inside the same country without ever being exported. This implies that for main producers, we have an error margin of the size of the domestic market consumption.

Exploratory data analysis. Before any visualization code was written, we ran a fairly extensive EDA in `EDA.ipynb`. Three findings shaped the dashboard. **(1)** The top raw-bean exporters (Brazil, Vietnam, Colombia, Indonesia, Ethiopia) dominate green-bean exports across the entire window but together export very little processed coffee. The producer/processor split is near-bimodal, not a continuum. **(2)** The major *processed*-coffee exporters (USA, Germany, Italy) collectively grow nearly no commercial coffee. They are entirely intermediary, the perfect Sankey middle column. **(3)** The ratio of average processed-coffee export price to average green-bean export price is large and persistent across the entire forty-year window, so the price-timeline view will tell a coherent story instead of dissolving into noise.

Pre-processing pipeline. The FAO trade matrix was downloaded from this [webpage](#); it included all traded agricultural commodities, not just coffee. Converting this heavy CSV file into a dataframe, we discarded rows whose `Item` column did not contain “coffee”, then narrowed the filtering to exclude coffee extracts, substitutes, husks, and skins. After targeting the right data, we removed redundant or irrelevant columns and dropped every year column that contained no data, as well as cleaning the source flags since only 1.3% was not “Official”. Finally, two preprocessed ready-to-use datasets were created: one filtering `Element Code` on import/export values (`Trade_Matrix_Value.csv`), the other on quantities (`Trade_Matrix_Quantity.csv`). From these two CSVs, a Node.js build step (`js/build_data.js`) harmonises country names against a world-atlas GeoJSON (via `REPORTER_CANONICAL` and `PARTNER_NORMALIZE` tables), aggregates per country per year, computes the processed-share metric needed by the Sankey, and emits a single compact `data_layer.json` file. This JSON is the only file the front-end ever loads at runtime. Because the file is several megabytes, it is hosted on a dedicated GitHub Pages site (mass-14.github.io/BeanMap-data/) maintained by one of the team members, so that the deployed Vercel front-end can fetch it cross-origin without bloating the main repository. The build step is idempotent: re-running it with updated CSVs simply overwrites the JSON on that Pages site.

3 The Evolution: From Flat Arrows to a 3D Story

The most important pivot happened during exploratory data analysis. We had built an early prototype of the flat-map-with-arrows idea, and it looked... fine. The problem was that it told the same story every other coffee visualization tells: “Brazil grows a lot of coffee, Europe drinks a lot of coffee, arrows go up.”

The key insight: mapping *quantities* alone misses the essential economic truth. The coffee bean undergoes a *physical transformation* (roasting, grinding, packaging into capsules) before reaching the consumer, and *the value resides in that processing stage*. The visualization has to encode **transformation**, not just trade.

Once we accepted that, several design choices fell out almost automatically. The Transformation Index became the primary visual variable. The map needed to encode a *magnitude* per country, not a flow between them, which is what 3D extrusion does well. And we needed a second view (a Sankey) to tell the dual flow story (*Producers* → *Processors* → *Consumers*) that the map alone cannot show.

From the Milestone 2 sketch to the final layout. Figure 1 shows the annotated interface design we produced for Milestone 2. That design was built around a *flat choropleth* map (region 2 in the annotation) shaded by the Transformation Index, with directional arrow overlays drawn between every visible pair of trading countries by default (region 3). A small statistics panel was pinned to the right (region 4), and the year timeline and Sankey were already present as lower-row views (regions 5–7).

That Milestone 2 layout had three structural problems that only became obvious once we started prototyping:

- **The arrow overlay was unreadable by default.** As soon as more than a few countries were displayed, arcs collided over the equatorial belt where most producers live. We fixed this in two stages: first gating arcs behind country selection (no selection means no arcs), then capping the visible arcs at top-2 exports and top-2 imports.
- **One channel was carrying too much load.** The choropleth could encode only one variable per country, so total trade volume took the slot and the more important question (transformation) had no visual predominance. The shift to 3D extrusion freed up height for TI specifically and let colour redundantly reinforce it, which made the producer/processor split immediately visible.

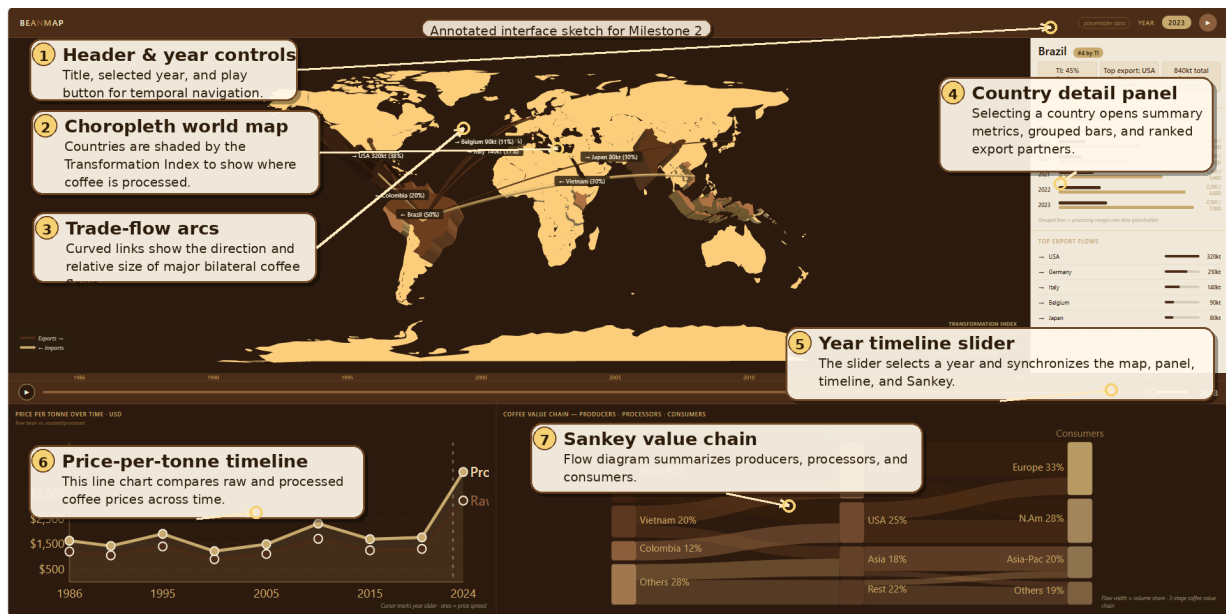


Figure 1: Annotated interface sketch produced for Milestone 2.

- **The story was static.** The Milestone 2 sketches were essentially a single-year snapshot. Adding the year slider, the price timeline, and the Sankey turned the dashboard from a snapshot into a four-decade movie, with three coordinated views that progressively disclose detail as the user interacts.

4 Design Decisions & Visual Encoding

The final dashboard (Figure 2) is driven by a single shared application state (current year, selected country, value/volume toggle). Each view earns its place by showing something the others cannot. The whole interface is themed to match the commodity: a palette of cream, latte and espresso provides coherence and avoids flashy colours that distract the user. Typography, font sizes and weights have also been calibrated to let the focus rest on the data rather than on styling.

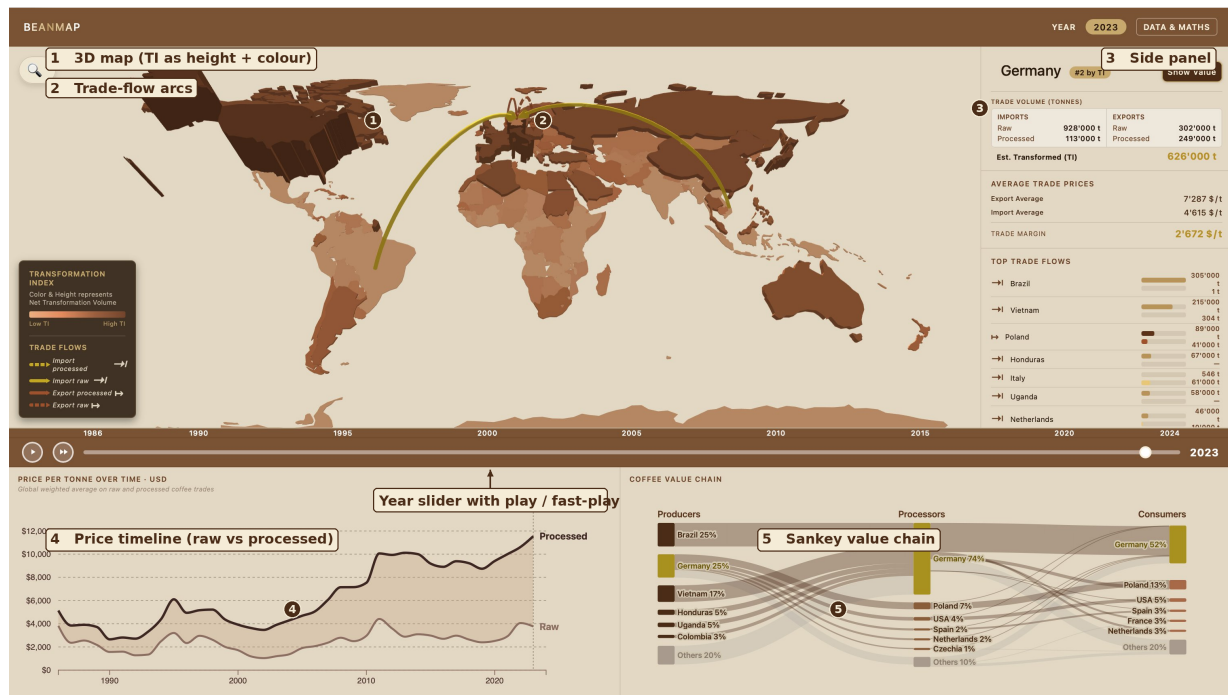


Figure 2: The BeanMap dashboard in its default state.

4.1 The 3D map

The centerpiece is a 3D extruded world map rendered on top of a world-atlas GeoJSON. Each country polygon is converted into an extruded prism whose depth and colour are both driven by its Transformation Index. We chose 3D in addition to the more conventional choropleth because height is a more salient channel than colour for encoding magnitude on a world map (the eye picks up a tall USA or Germany in a fraction of a second, even for small countries) and because the 3D scene affords rotation, zoom, and orbit interactions that turn passive reading into exploration.

Marks and channels. Two visual channels encode the same quantity (TI). **Height** is the primary channel: extrusion depth is proportional to $TI/TI_{\max}$, where $TI_{\max}$ is a *global* maximum computed once across all years. Using a global anchor (not a per-year one) means heights are directly comparable across years, so growing pillars genuinely correspond to growing transformation when the slider is scrubbed. **Colour** is the redundant channel: a four-stop coffee-toned gradient interpolated with a low-exponent power curve. The exponent is critical: the TI distribution is heavily right-skewed (a handful of giant processors and a long tail), so a linear ramp would compress everything below the top into a near-uniform pale region. The power curve stretches the bottom so that mid-tier processors become visually distinguishable from non-processors. Encoding TI redundantly on height and colour is also an accessibility choice: users who cannot easily judge 3D depth still get the magnitude through colour, and vice versa.

Camera, lighting, and trackpad pan. The orbit controls are deliberately constrained, so that the user cannot flip the map upside-down or lose orientation. Country meshes are lit by a warm directional key plus a tan fill, which produces a smooth natural feel that echoes the colour palette. Trackpad two-finger pans are distinguished from mouse-wheel and pinch zoom by a custom listener, so that trackpad scrolling pans the camera over the map rather than zooming in and out, which is the behaviour the average laptop user expects.

4.2 Trade-flow arcs

On selection, the map overlays the top-2 export destinations and top-2 import sources for that country (Figure 3). We capped at two of each, not top- N , because trade is heavily concentrated and the equatorial belt became unreadable with more arcs. The arcs encode four pieces of information simultaneously:

Hue encodes direction: imports in yellow, exports in brown;

Stroke pattern encodes whether the dominant share of that bilateral flow is raw or processed, with dashed strokes reserved for the “non money-making” case in each direction (raw exports and processed imports), drawing the eye to the bilateral relationships that defy the producer/processor split;

Width encodes magnitude, scaled by $\sqrt{q/q_{\max}}$ to compress the order-of-magnitude range without losing rank order;

Vertical curvature encodes geographic distance, so long-haul transatlantic flows arch high above the scene while intra-European flows skim the surface, which keeps overlapping arcs from drawing on top of one another.

A toggle (**Show Value** / **Show Volume**) switches arcs and side-panel numbers between tonnes and USD; both views are computed from the same source, but the same flow can tell materially different stories in the two modes.

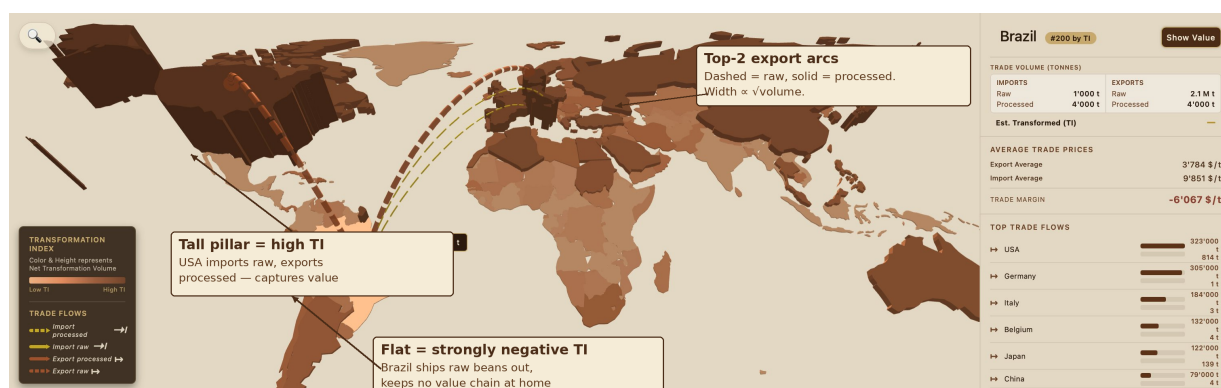


Figure 3: Brazil selected, 2023.

4.3 Sankey value chain

The Sankey (Figure 4) shows three columns: **Producers** → **Processors** → **Consumers**. A country is classified as a Processor if at least **30%** of its coffee exports are roasted. The 30% cut keeps the middle column clean: at lower thresholds it would be polluted with producing countries that occasionally re-export a small amount

of roasted coffee; at higher thresholds legitimate hubs would disappear. The diagram shows the top eight countries in each column, dropping to six when a country is selected to leave room for the focus country, which is force-included.

Mass balance. The Sankey is strictly mass-conserving. For every active country we compute a “retained” flow: raw coffee imported but never re-exported in any form, the residual that by mass balance must have been consumed domestically, and add it to the Processors → Consumers band of the same country, so the diagram’s left edge (raw exports plus retained) exactly balances its right edge (total consumed). Without this band, mass would silently leak out and the producing world’s contribution would appear smaller than it really is.

Selection-aware re-layout. When a country is selected on the map, the Sankey re-runs its layout with that country forced into each column where it appears, and with all flows that don’t touch the selection filtered out. Clicking “Italy” on the map gives you the Sankey of Italy’s specific processor-to-consumer relationships (Figure 4), not the global Sankey with Italy highlighted, which is a much stronger insight into the weight Italy carries in the coffee chain than mere highlighting would give.

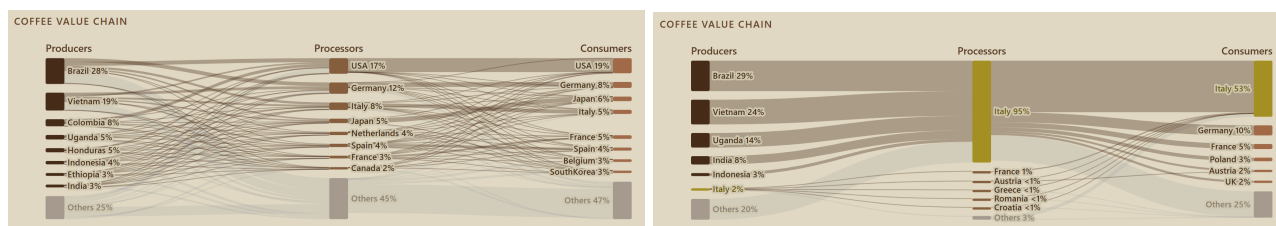


Figure 4: Sankey value chain diagram for 2023, with no selected country (left) and with Italy selected (right).

4.4 Price timeline

A D3 line chart (Figure 5) plots the global weighted average price per tonne in USD for raw and processed coffee from 1986 to the selected year, with a shaded area between the two lines. The lines are weighted averages (total world trade value divided by total world trade volume in that year) which reflects the prices at which most coffee actually changed hands rather than over-weighting atypical flows. A dashed vertical cursor tracks the current year, and a focus-line overlay exposes exact numerical values on hover.

The gap between raw and processed lines is the value-added gap, and watching it widen and narrow over four decades is one of the most evocative parts of the dashboard. Price shocks are absorbed differently at each end of the chain: the raw-bean line spikes visibly on major supply disruptions, while the processed line remains markedly smoother, which is itself a finding. The processing economy absorbs commodity volatility and is rewarded for that with a stable margin.

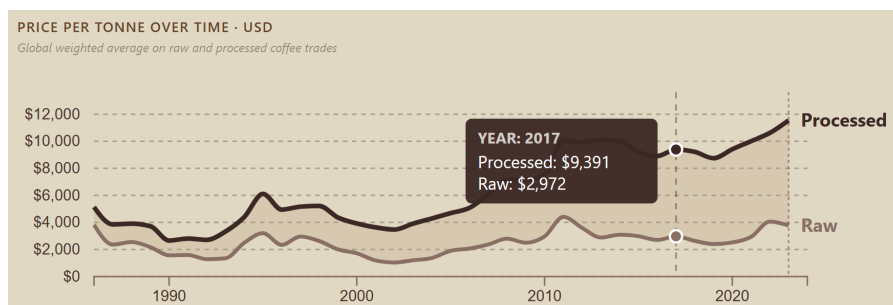


Figure 5: Price per tonne over time 1986–2024, with a hover on year 2017.

4.5 Side panel, year slider, search

The side panel (visible on the right of Figure 3) summarises all the data for the selected country:

A header with country name and a producer/processor rank badge;

A 2×2 grid of imports/exports broken into raw/processed, expressed in tonnes or USD depending on the toggle;

The estimated transformed-volume figure (TV) with the estimated trade margin (added-value) made by the country;

And a scrollable ranked list of top trading partners. The information is duplicated from the arcs and Sankey on purpose, so the user gets both the full list of partners and the numerical details.

The year slider has a normal-speed play button and a fast-play button. The normal speed is useful for presentation, the fast one for evolution exploration. Pressing **Escape** clears the selection. A search box with autocomplete gives keyboard access to any country, which is especially useful for small countries whose polygons are hard to click on the map.

4.6 The “Data & Maths” panel

A button in the top-right of the interface opens a dedicated **Data & Maths** panel that documents, in the user’s own session, exactly how every number on the screen was computed. It lays out the Transformation Index formula, the mass-balance equation that drives the Sankey’s retained band, the weighted-average computation behind the price timeline, the data source and item codes, and a glossary of every label that appears in the dashboard. We added this panel late in the design cycle after realising that a viewer who is shown a 3D pillar labelled “Italy” is owed a clear explanation of what that pillar actually measures, otherwise the dashboard looks confident but unaccountable.

5 Challenges & Technical Implementation

5.1 Architecture

The front-end is a static single-page app with no build step, which keeps deployment trivial: it ships on Vercel as plain HTML/CSS/JS with zero server-side dependencies. The code is split into seven focused JavaScript modules plus an offline pre-processor (`app.js`, `config.js`, `dataService.js`, `map3D.js`, `visualizations.js`, `uiController.js`, `utils.js`, and `build_data.js`). The split is functional, not arbitrary: each module owns one part of the dashboard, all view modules subscribe to the same shared application state, and no view module reaches into another’s DOM.

5.2 Library stack

The runtime stack is deliberately small. We pull only three external libraries: **Three.js** for the 3D scene; **D3** for the price timeline and DOM binding; **d3-sankey** as a standalone plugin to keep page-load weight down. We deliberately did *not* pull in a front-end framework (React, Vue, Svelte). The dashboard is single-page and single-user, with a small shared mutable state, which is a workload a framework would handle, but at the cost of significant bundle size, a build step, and a deployment pipeline. The hand-rolled approach does the same job in a few dozen lines and has the advantage of being trivially debuggable.

5.3 Performance: pre-baking and diff-based rendering

Making the dashboard feel fluid while it reasons over forty years of bilateral trade took three design decisions. **Pre-baking:** heavy aggregation happens once, offline, in the build step; the runtime reads a single compact JSON pre-indexed by year, so every view looks up its data in $O(1)$. **Diff-based re-rendering:** when the year slider moves, the country meshes update their colour and height in place rather than being rebuilt; the Sankey re-runs its layout only when the year or selection changes; the price timeline progressively reveals data points rather than redrawing from scratch. **Cheap shading:** the 3D map uses the lightest non-flat-shaded material available, which is more than enough for a map. The net effect is that dragging the slider produces real-time updates of around 200 country meshes plus a handful of arcs, well within 60 fps on modest laptops.

5.4 Geographic and reporting quirks

A surprising amount of work went into reconciling country names. The trade matrix, the GeoJSON, and natural language disagree about “Czechia” vs. “Czech Republic”, former-USSR successor states, the two Congos, “Côte d’Ivoire” vs. “Ivory Coast”, and “Belgium-Luxembourg” (a single reporter until 1999). We resolved these in two layers: the pre-processor canonicalises names at build time, and a runtime alias table maps the GeoJSON’s geographic labels to the same canonical keys. A second class of quirk: not every country reports in every year. We treated missing data as *absent*, not zero. A country with no reported trade in 1992 simply does not extrude that year, rather than being silently mislabelled as a producer of zero coffee.

5.5 The turning point

The breakthrough came when we hooked up the first coordinated view: the 3D map and the synchronised Sankey, both driven by the same year slider. As soon as we dragged the slider from 1986 to 2024, we could literally watch Brazil and Vietnam stay flat at the bottom of the world while the USA, Germany, Belgium and Italy *grew*, decade by decade, into tall pillars of value extraction. That single animation convinced us TI was the right metric, and everything else was downstream.

6 Final Product Showcase

Insights visible at a glance.

- **Italy is taller than Brazil.** A country that grows only 2% of the coffee flowing through it rises to one of the tallest extrusions on the map, because it imports raw beans from Brazil, Vietnam, Uganda and others,

roasts and packages them domestically, and re-exports processed coffee across Europe. This one comparison, available with a single mouse-hover, is the whole thesis of the project in one image.

- **The producing world barely roasts.** Of the top producing countries, only a small handful export meaningful volumes of *roasted* coffee. The Sankey shows the producers' band collapsing almost entirely into the processors' band before fanning out to consumers. So we can assume the majority of producers only process the domestic-market coffee.
- **The value-added gap is widening.** Raw and processed price lines diverge over the forty-year window, especially since the early 2000s.
- **Volatility lives upstream.** The raw-price line is far jumpier than the processed-price line. The processing economy absorbs commodity shocks on behalf of the consumer and is rewarded for that with a stable margin.

Information density and progressive disclosure. The dashboard is dense by design, with five coordinated views on a single page, but we manage that density with progressive disclosure. On first load, Germany is pre-selected and 2023 pre-set, so the user lands on a fully-populated, story-telling view instead of a blank slate. Hovering reveals tooltips; clicking re-populates the side panel, redraws the arcs, and re-filters the Sankey around the selection; the slider and the value/volume toggle act on the current selection. Every interaction adds detail without replacing the view, so the user can always read the global story *and* the local one at the same time.

What we would do with more time. Three directions. First, add a layer for *consumer retail prices* (supermarket-shelf or café-cup), which would close the loop on the bean's journey and reveal the second great margin: a roasted-coffee export price of ~\$5–7/kg vs. a retail espresso equivalent to \$100+/kg of dry coffee. Second, push the data back to 1962 using older releases to include the 1989 collapse of the International Coffee Agreement, a structural break the dashboard currently has to tell from the post-collapse equilibrium onward. Third, a *differential* mode that contrasts two years side by side, for users who do not want to scrub the slider themselves.

7 Conclusion & Peer Assessment

Building BeanMap taught us that the hardest part of a data visualization is rarely the rendering, it is finding the right *metric*. Our first prototype, the flat map with quantity arrows, was technically correct and narratively dull. The shift to the Transformation Index reframed the same underlying data into a story about who actually captures value in the coffee economy, and everything else (the 3D extrusion, the Sankey, the coordinated year slider) was downstream of that decision. The project also gave us a concrete laboratory for the course's core concepts: marks-and-channels reasoning (height vs. colour vs. width, and the use of a power-curve scale to handle a skewed distribution), interaction design (linked views, brushed selections, scrubbable time), and narrative scaffolding (a single page readable at three different time scales). A second, less obvious lesson is that *honest* encoding matters more than dramatic encoding: the mass-balance Sankey, the globally-anchored TI normalisation, the choice to treat missing data as absent rather than zero, none is visually striking on its own, but together they make the dashboard trustworthy.

Peer assessment

Team member	Main contributions
Noam Ifergan	Designed and coded the full prototype until the final stage complete MVP.
Massimo Berardi	EDA notebook, from MVP to final website (single-file MVP into a production-standards modular codebase, bug and logic fixing, colour and rendering rework, user-test).
Victor Nahoul	Wrote the process book from storytelling to implementation details, screencast producer and narrator.

All three team members contributed jointly to the original idea, the dataset research, the design pivots (especially the flat-to-3D shift and the choice of TI as the central metric), the iterative review of every component in the final dashboard, the writing of the two first milestones' reports, and the review of the process book. The task distribution was mainly driven by personal interests, alongside skills and availability, allowing each member to work without stepping on the others.